

# FMAM 2026 rainfall outlook

ECMWF SEAS5 initialized 1 January 2026 · calibrated with CHIRPS v2.0 · FMAM rainfall domain (FMAM  $\geq$  20% of annual and  $\geq$  100 mm)

## Research reconstruction — not an official EMI or ICPAC forecast

The forecast is close to climatology in the FMAM rainfall domain (FMAM  $\geq$  20% of annual and  $\geq$  100 mm): no tercile stands out on average. Mean forecast rainfall is 284 mm against a 1993–2025 average of 287 mm (−3 mm, −1%).



**Below normal 35%    Near normal 32%    Above normal 34%**

Average local probability: each grid cell has its own tercile probabilities; these are their averages over the 100% of the area that has probabilities, not the probability that the area-total rainfall falls in a category.

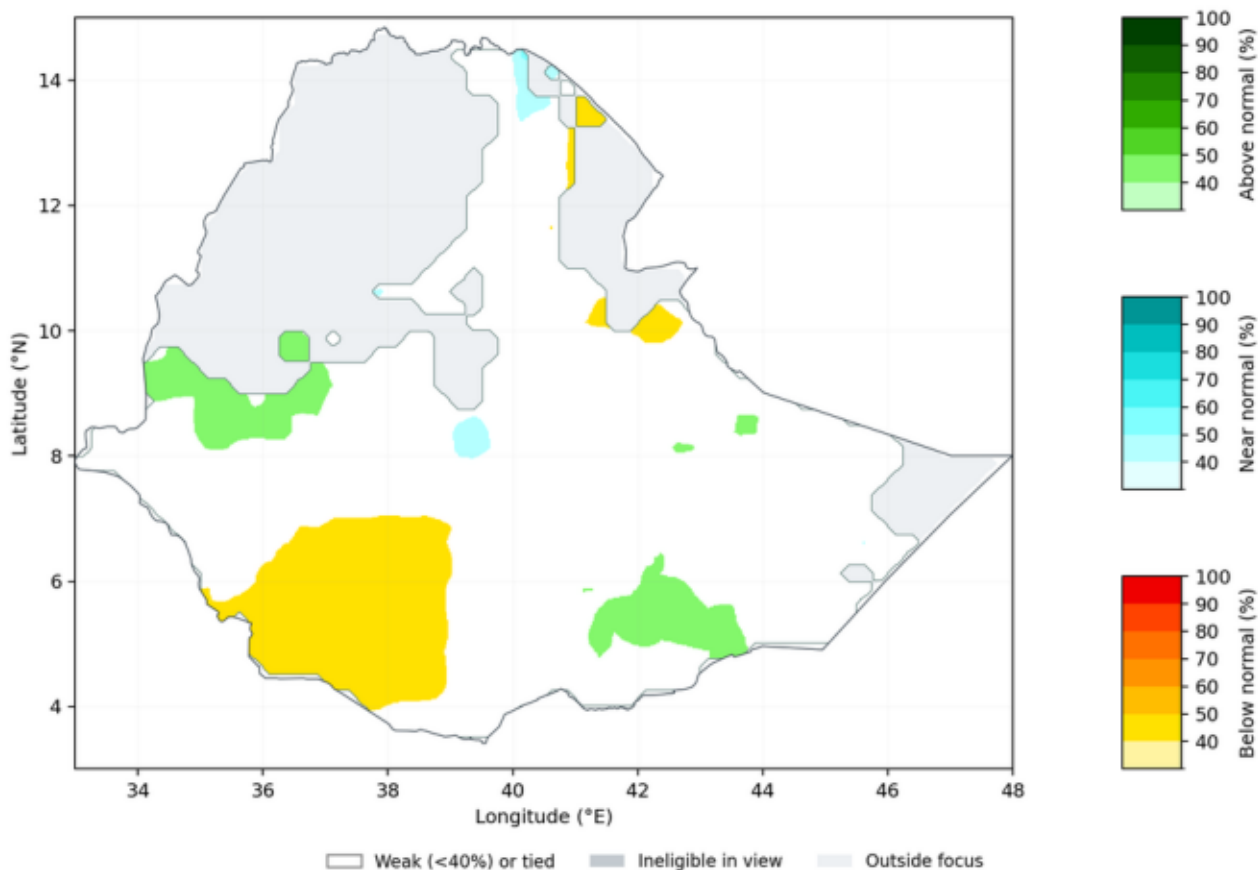
Forecast signal: No clear tilt (close to climatology) (largest area-average probability compared with a one-third reference; based on probabilities only).

Historical skill (FMAM rainfall domain (FMAM  $\geq$  20% of annual and  $\geq$  100 mm)): Small estimated improvement; skill uncertain. Probability skill (RPSS) +0.031, cross-validated 1993–2016, 95% interval −0.012 to +0.069; better than climatology in 17 of 24 years. A strong signal is not a confidence rating.

## FMAM 2026 | rainfall tercile outlook

January initialization · shared probability blend · CHIRPS reference 1993–2025

FMAM rainfall domain (FMAM  $\geq$  20% of annual and  $\geq$  100 mm) · presentation layer



Display interpolation: 12x; Gaussian  $\sigma=0.6$  native cells. Statistics and exported fields use the original 0.25° grid.  
Research reconstruction of a January-initialized forecast; not an official EMI/ICPAC product.

# All targets, evidence and record

FMAM 2026 — January initialization · FMAM rainfall domain (FMAM >= 20% of annual and >= 100 mm)

| Target    | Below/near/above | Coverage | Anomaly | Verification status    |
|-----------|------------------|----------|---------|------------------------|
| FMAM 2026 | 35% / 32% / 34%  | 100%     | −3 mm   | Verification published |
| Feb 2026  | 28% / 40% / 33%  | 81%      | +1 mm   | Verification published |
| Mar 2026  | 31% / 35% / 34%  | 91%      | +7 mm   | Verification published |
| Apr 2026  | 36% / 31% / 33%  | 100%     | −4 mm   | Verification published |
| May 2026  | 36% / 34% / 30%  | 100%     | −6 mm   | Verification published |

## Historical skill, cross-validated (probability skill, RPSS, with whole-year 95% interval)

| Target | RPSS (95% interval)       | Yrs better | Holm p | In words                                                  |
|--------|---------------------------|------------|--------|-----------------------------------------------------------|
| FMAM   | +0.031 (−0.012 to +0.069) | 17/24      | 0.249  | Small estimated improvement; skill uncertain              |
| Feb    | +0.036 (0.000 to +0.072)  | 17/24      | 0.123  | Modest improvement over climatology (interval above zero) |
| Mar    | −0.001 (−0.018 to +0.015) | 17/24      | 1.000  | No demonstrated improvement over climatology              |
| Apr    | −0.002 (−0.025 to +0.018) | 13/24      | 1.000  | No demonstrated improvement over climatology              |
| May    | +0.034 (+0.004 to +0.061) | 19/24      | 0.084  | Modest improvement over climatology (interval above zero) |

## Verification against CHIRPS (single season or month)

| Target    | RPSS   | CRPSS  | Bias     | Forecast / observed anomaly | Observed below/near/above |
|-----------|--------|--------|----------|-----------------------------|---------------------------|
| FMAM 2026 | −0.037 | −0.071 | −58.7 mm | −3 / +56 mm                 | 3% / 45% / 52%            |
| Feb 2026  | −0.006 | −0.131 | +0.2 mm  | +1 / 0 mm                   | 32% / 44% / 24%           |
| Mar 2026  | +0.040 | +0.089 | −69.2 mm | +7 / +77 mm                 | 0% / 2% / 98%             |
| Apr 2026  | +0.035 | +0.050 | +11.7 mm | −4 / −16 mm                 | 44% / 37% / 19%           |
| May 2026  | +0.014 | −0.025 | −1.0 mm  | −6 / −5 mm                  | 38% / 44% / 18%           |

## Cycle record

Model initialization: 1 January 2026 · ECMWF SEAS5 (system 51), 51 members

Target period: 1 Feb 2026 – 31 May 2026

Forecast record: Frozen 8 Oct 2026 (after the target period ended, before this project downloaded its observations; SHA-256 manifest)

Publication status: Research reconstruction: produced after the initialization date with the published method; not an official EMI or ICPAC forecast

Reference period: CHIRPS v2.0 FMAM 1993–2025 (33 seasons)

Observations processed through: May 2026 (Feb, Mar, Apr, May)

CHIRPS archive: Official CHIRPS v2.0 p25 by\_month listing checked 10 Oct 2026: monthly files through Aug 2026; Feb 2026 file not found

Verification last run: 10 Oct 2026

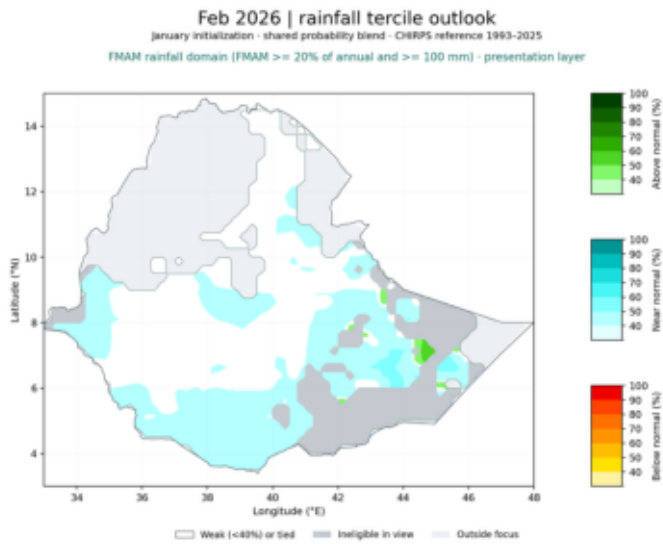
Method / version: Shared climatology blend ( $\lambda = 0.43$  for FMAM) · code 535e338

How to read: terciles split the reference-period rainfall into three equally likely parts. Climatology assigns about one-third to each (the observed tercile frequencies of the training years). Skill scores are decimals: +0.05 means a 5% lower score than climatology. RPSS refers to the blended probabilities; CRPSS and bias to the amount-corrected ensemble.

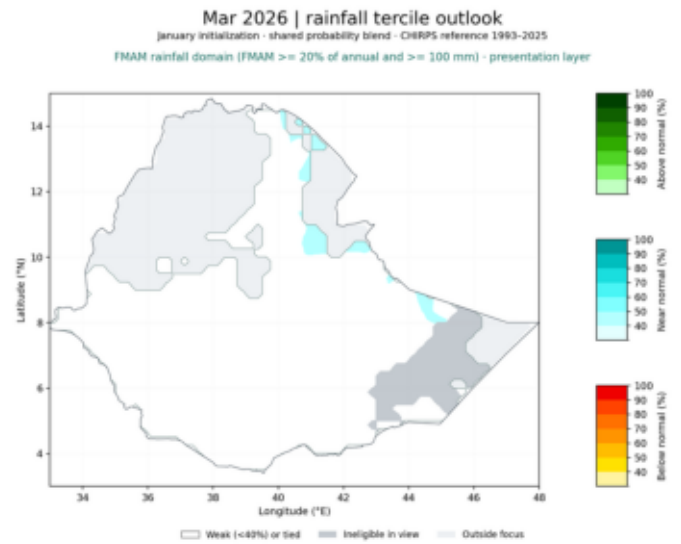
Limitations: few evaluation years and single verification seasons give wide uncertainty; the 0.25° maps repeat ~1° model information; the rainfall domain is for display and summaries only.

# Monthly tercile outlooks

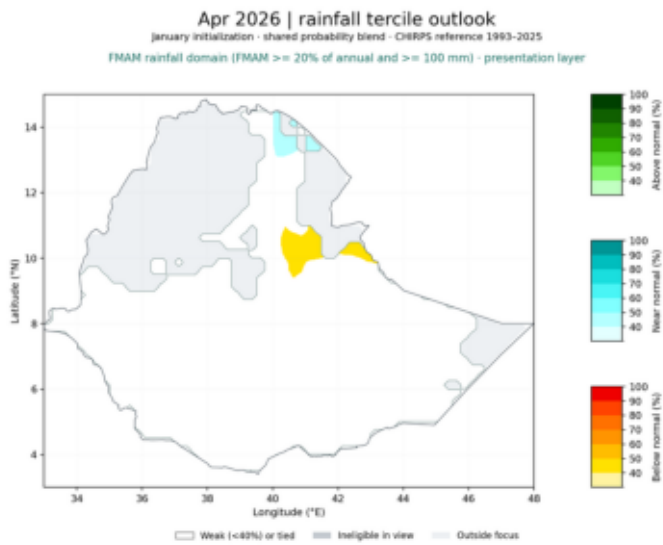
FMAM 2026 — January initialization · FMAM rainfall domain (FMAM  $\geq 20\%$  of annual and  $\geq 100$  mm)



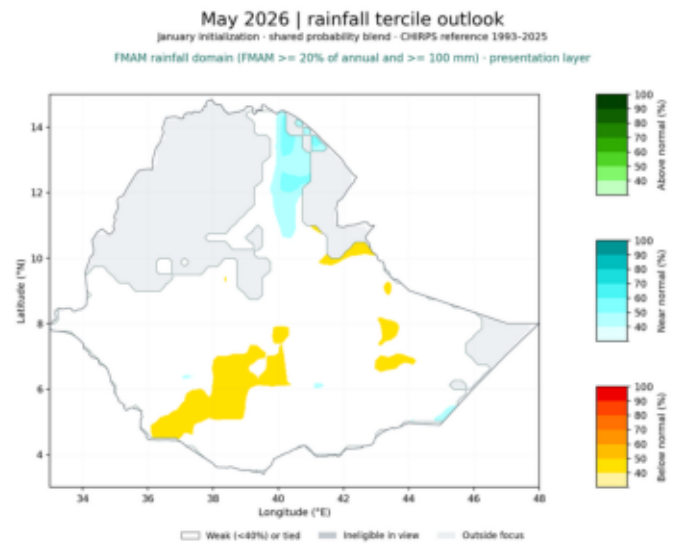
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